CITY
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Week 9
Systems Security

IN 1011 Operating Systems

Systems Security

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- Computer Security Overview
- Problems: Security Violation Methods
- Actions to Mitigate the Problems
- Authentication, Passwords and Alternatives
- More on Threats

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Protection ≠ Security

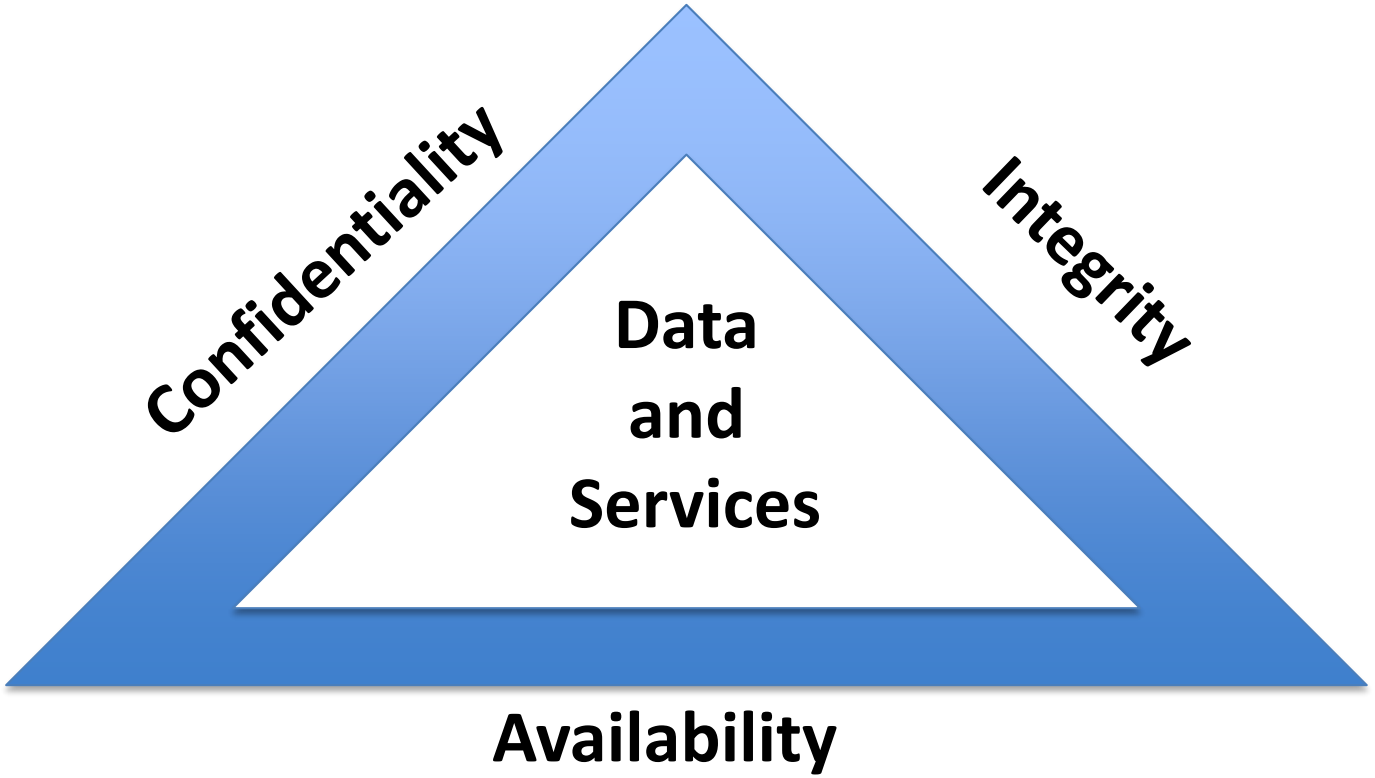
- File system protection mechanisms may prevent a process owned by `watson` from accessing a file owned by `sherlock`.
- But what if `watson` can login as `sherlock` (masquerading)?
  - Then `watson` can execute processes owned by `sherlock`
  - So `watson` can see and modify `sherlock's` data and processes

Computer Security Overview

- The National Institute of Standards and Technology (NIST https://www.nist.gov/) Computer Security Handbook 1995 (https://www.nist.gov/publications/introduction-computer-security-nist-handbook) defines the term Computer Security as:
  - "The protection afforded to an automated information system in order to attain the applicable objectives of preserving the **integrity**, **availability** and **confidentiality** of information system resources"

Computer Security Overview

– Preserving the **integrity**, **availability** and **confidentiality** is sometimes known as the **CIA Triad**.



Key Security Concepts

- **Confidentiality**
  - Preserving authorized restrictions on information access and disclosure, including means for protecting personal privacy and proprietary information
- **Integrity**
  - Guarding against improper information modification or destruction, including ensuring information nonrepudiation and authenticity
- **Availability**
  - Ensuring timely and reliable access to and use of information

The Security Problem

- System secure if resources used and accessed as intended under all circumstances
  - Unachievable
- **Intruders** (crackers) attempt to breach security
- **Threat** is potential security violation
- **Attack** is attempt to breach security (which can be accidental or malicious)
- Easier to protect against accidental than malicious misuse

Security Violations

- Breach of confidentiality
  - Unauthorized reading of data
- Breach of integrity
  - Unauthorized modification of data
- Breach of availability
  - Unauthorized destruction of data
- Theft of service
  - Unauthorized use of resources
- Denial of service (DOS)
  - Prevention of legitimate use

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Some Computer Security Challenges (1)

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- Computer security is not as simple as it might first appear to the novice
- Potential **attacks on the security features** must be considered
- Procedures used to provide particular services are often counterintuitive
- **Physical and logical placement** needs to be determined
- **Additional algorithms** or protocols are usually needed on top of what may be provided by the functional requirements of the system

Some Computer Security Challenges (2)

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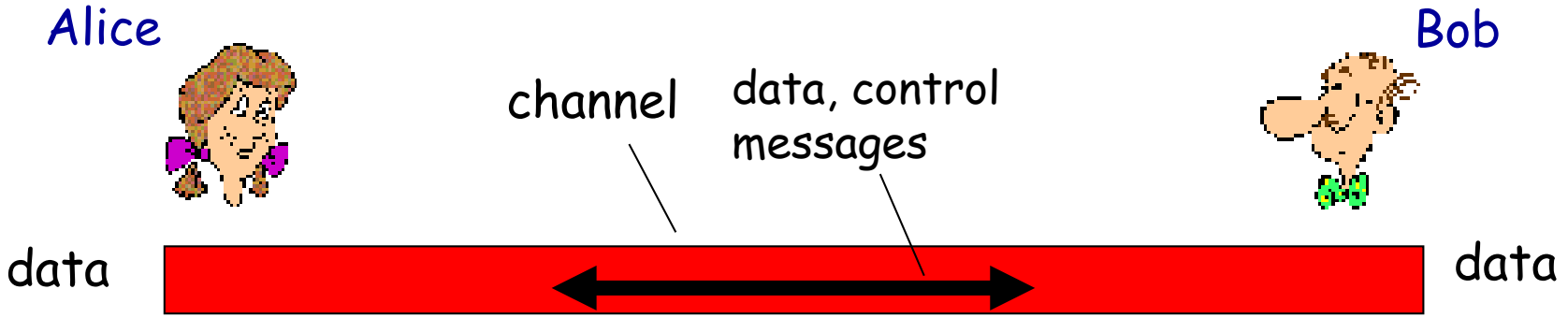
- Attackers only need to find **a single weakness**, the developer needs to find **all weaknesses**
- Users and system managers tend to not see the benefits of security until a failure occurs
- Security requires **regular and constant monitoring**
- Security is often an **afterthought** incorporated into a system after the design is complete
- Thought of as an **impediment** to efficient and user-friendly operation

Friends and enemies: Alice, Bob, …

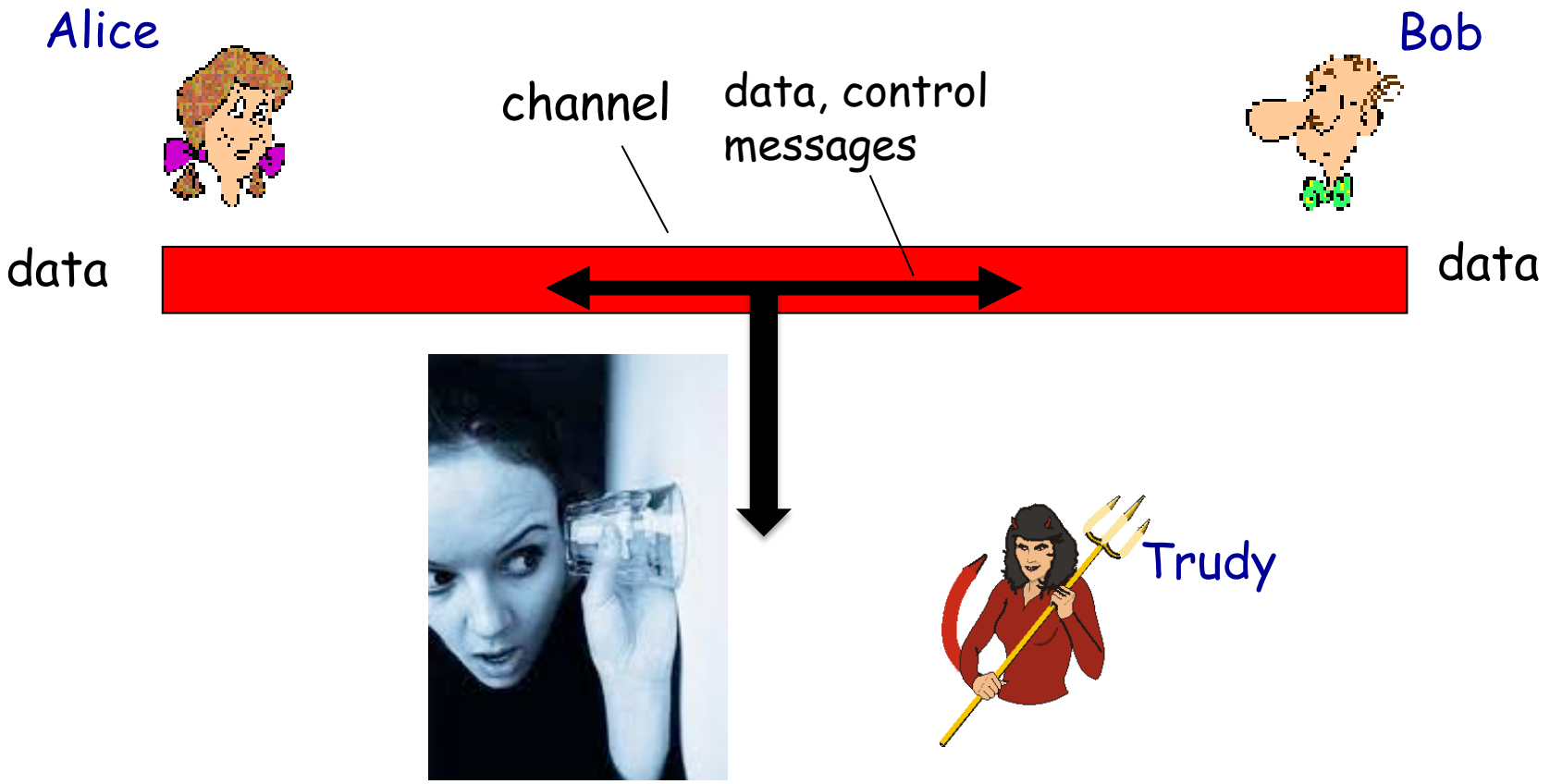
- Well-known in network security world after Rivest classic RSA paper (https://dl.acm.org/doi/10.1145/359340.359342).



- *Carol*, *Dan* third / fourth participant
- *Chuck*, a participant usually of malicious intent, *Craig*, cracker
- *Eve*, an eavesdropper
- *Mallory* or *Mallet*, a malicious attacker
- *Trudy*, an intruder.
- *Trent*, a trusted arbitrator, …

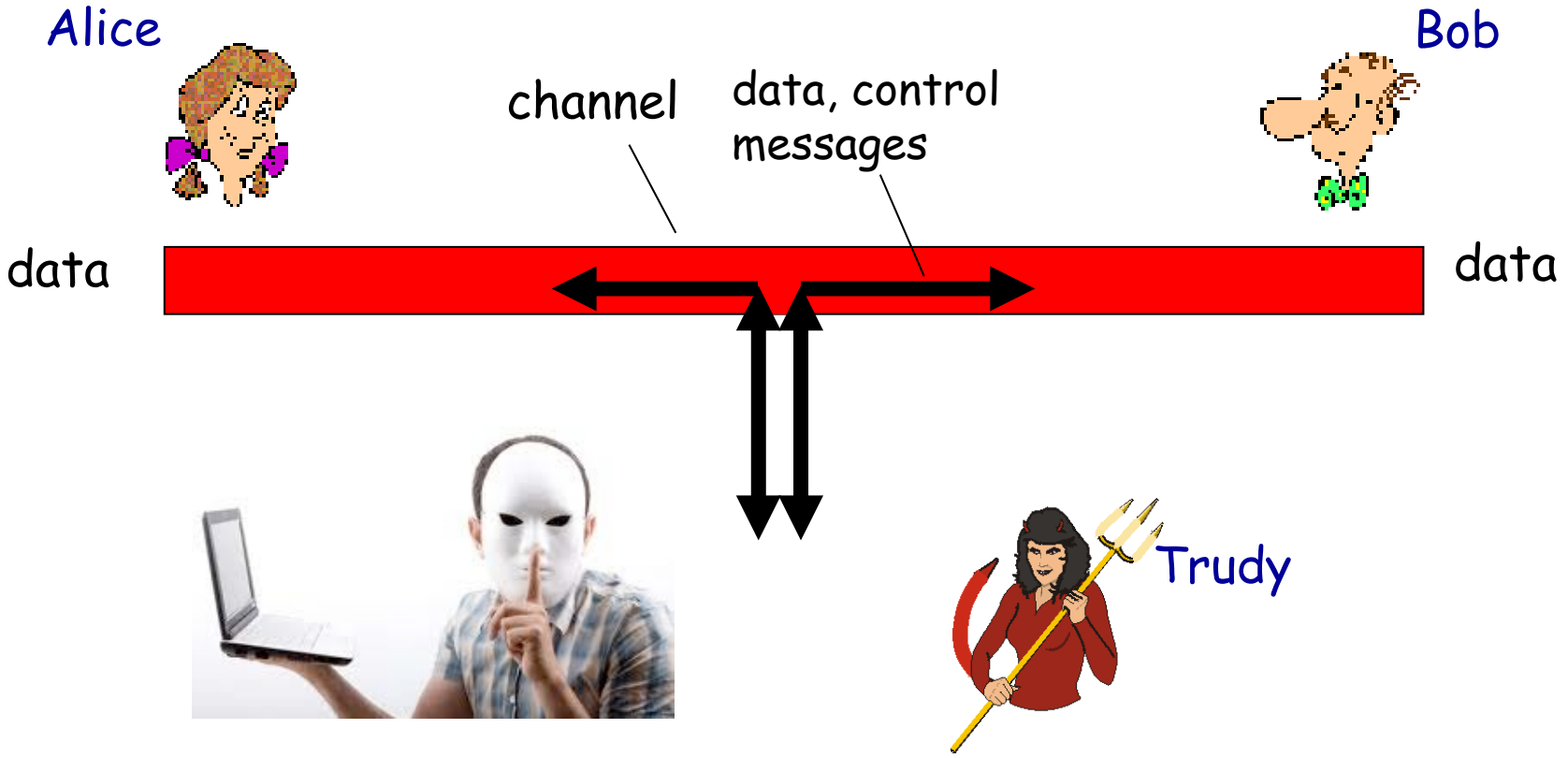
Potential problems for Bob and Alice (Security Violation Methods)

- Eavesdroppers in the conversation



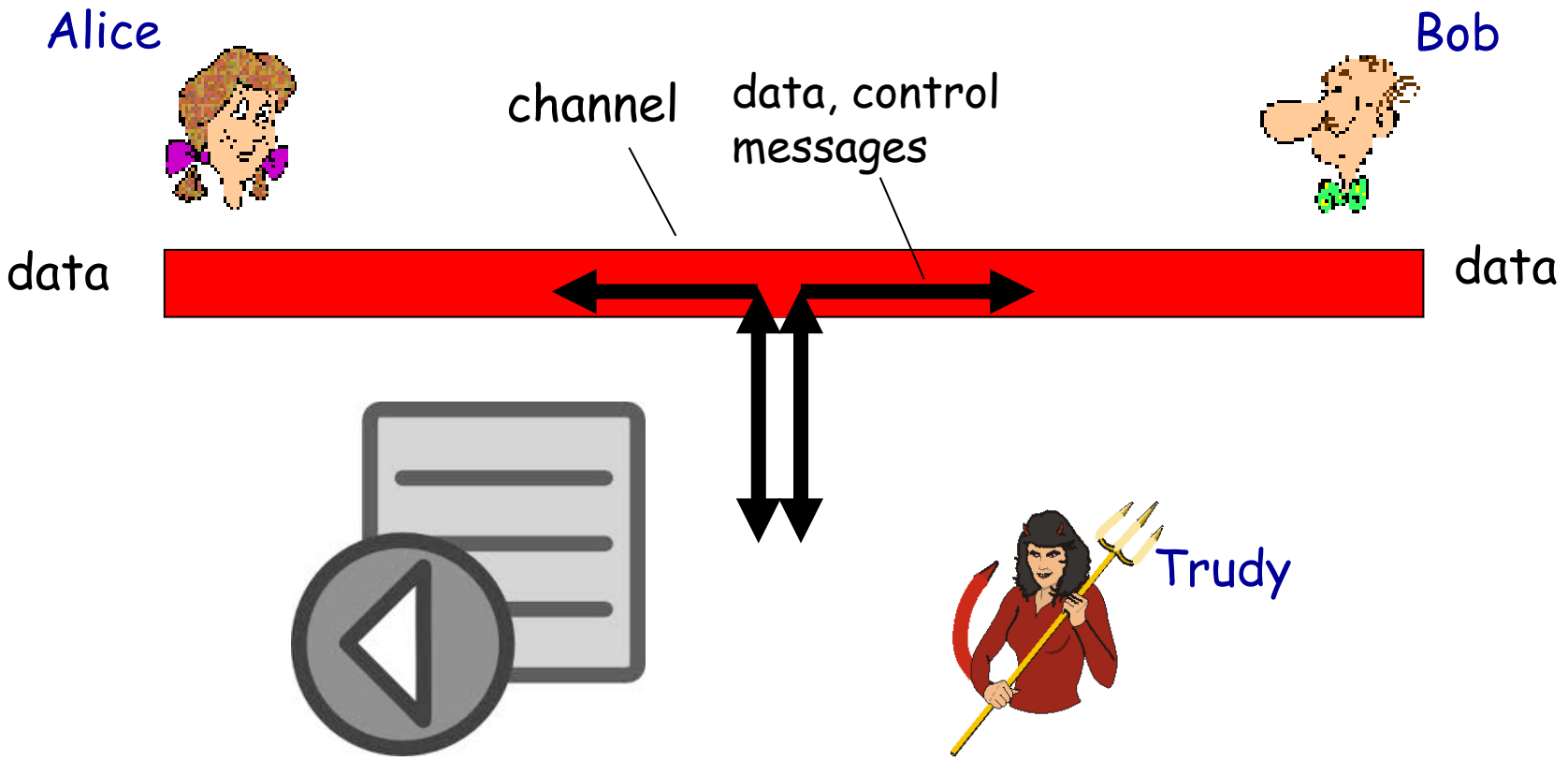
Potential problems for Bob and Alice (Security Violation Methods)

- Impersonation of users, Masquerading breach authentication



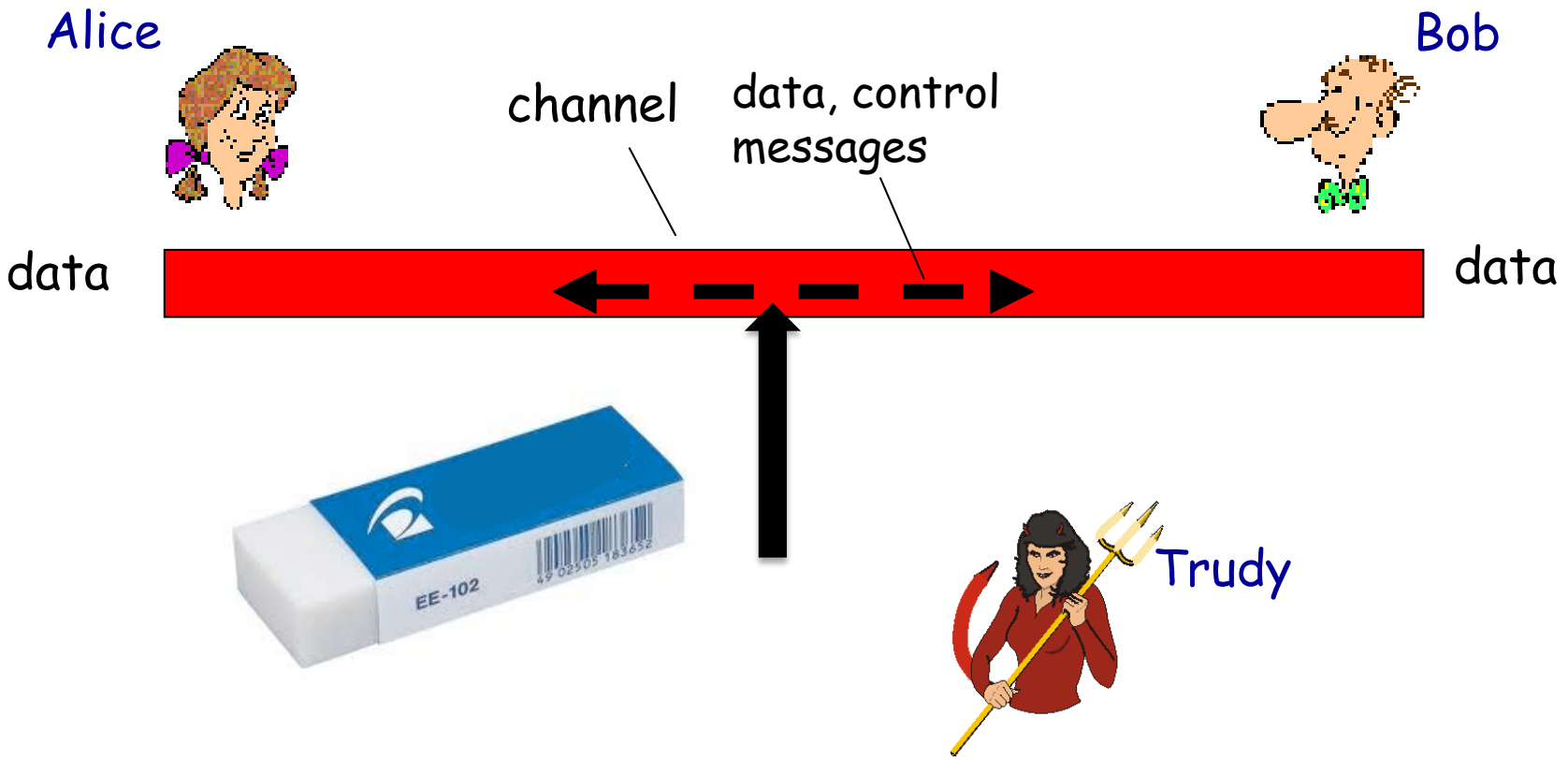
Potential problems for Bob and Alice (Security Violation Methods)

- Re-play of genuine / bogus messages



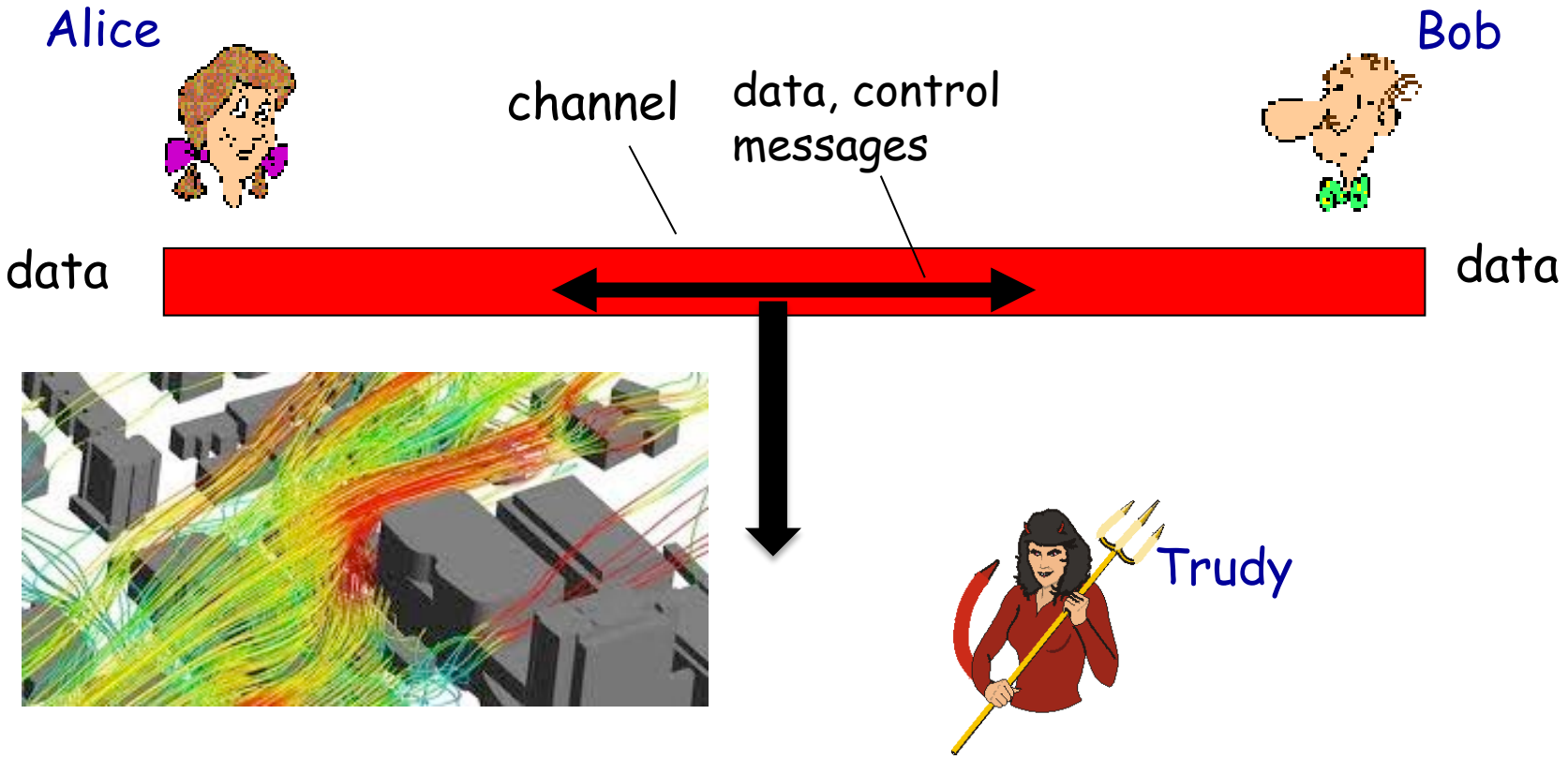
Potential problems for Bob and Alice (Security Violation Methods)

- Deleting or modifying messages



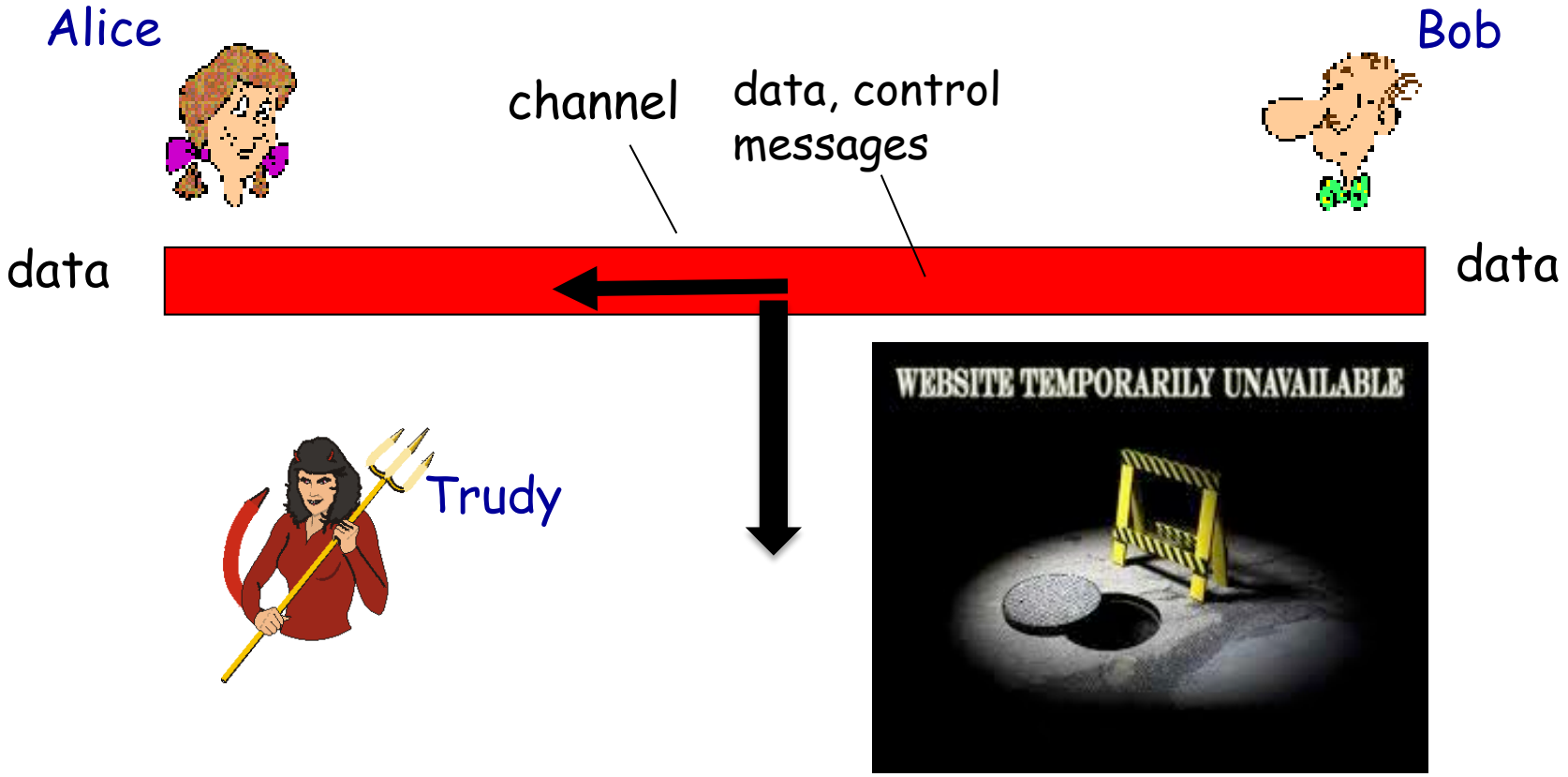
Potential problems for Bob and Alice (Security Violation Methods)

- Analysing the flow



Potential problems for Bob and Alice (Security Violation Methods)

- Blocking / diverting



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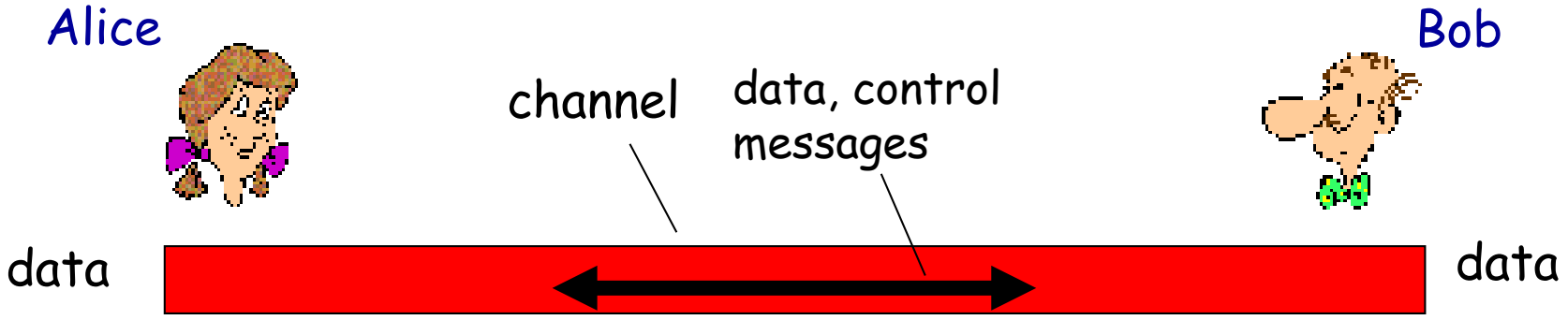
Security Measure Levels

- Impossible to have absolute security, but make cost to perpetrator sufficiently high to deter most intruders
- Security must occur at four levels to be effective:
  - Physical
    - Data centers, servers, connected terminals
  - Human
    - Avoid social engineering, phishing, dumpster diving
  - Operating System
    - Protection mechanisms, debugging
  - Network
    - Intercepted communications, interruption, DOS
- Security is as weak as the weakest link in the chain
- But can too much security be a problem?

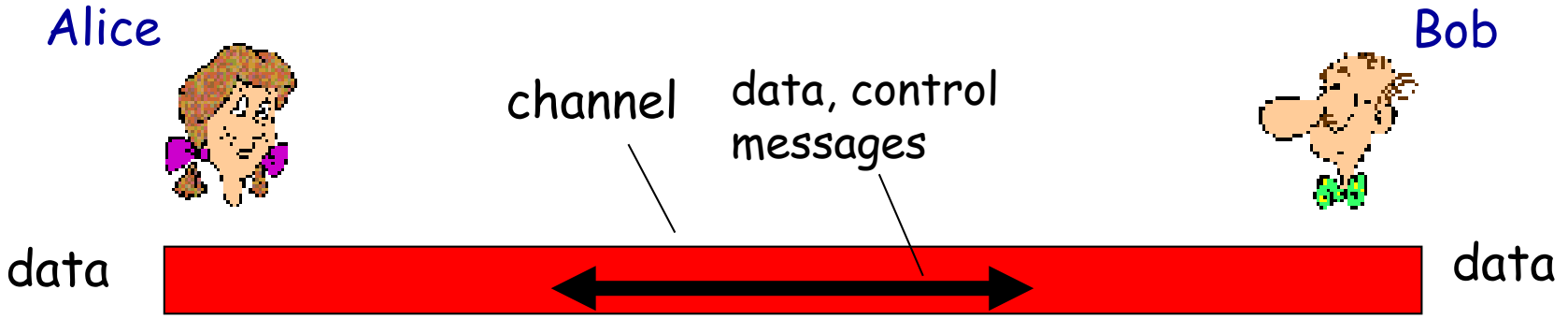
Potential Actions by Alice and Bob

- Authentication: sender, receiver want to confirm identity of each other
- RFC (request for comments) 2828 defines user authentication as:
  - *The process of verifying an identity claimed by or for a system entity.*



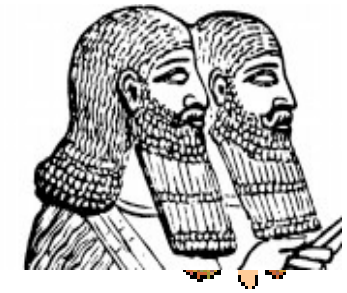
Potential Actions by Alice and Bob

- Message integrity: sender, receiver want to ensure message not altered (in transit, or afterwards) without detection.



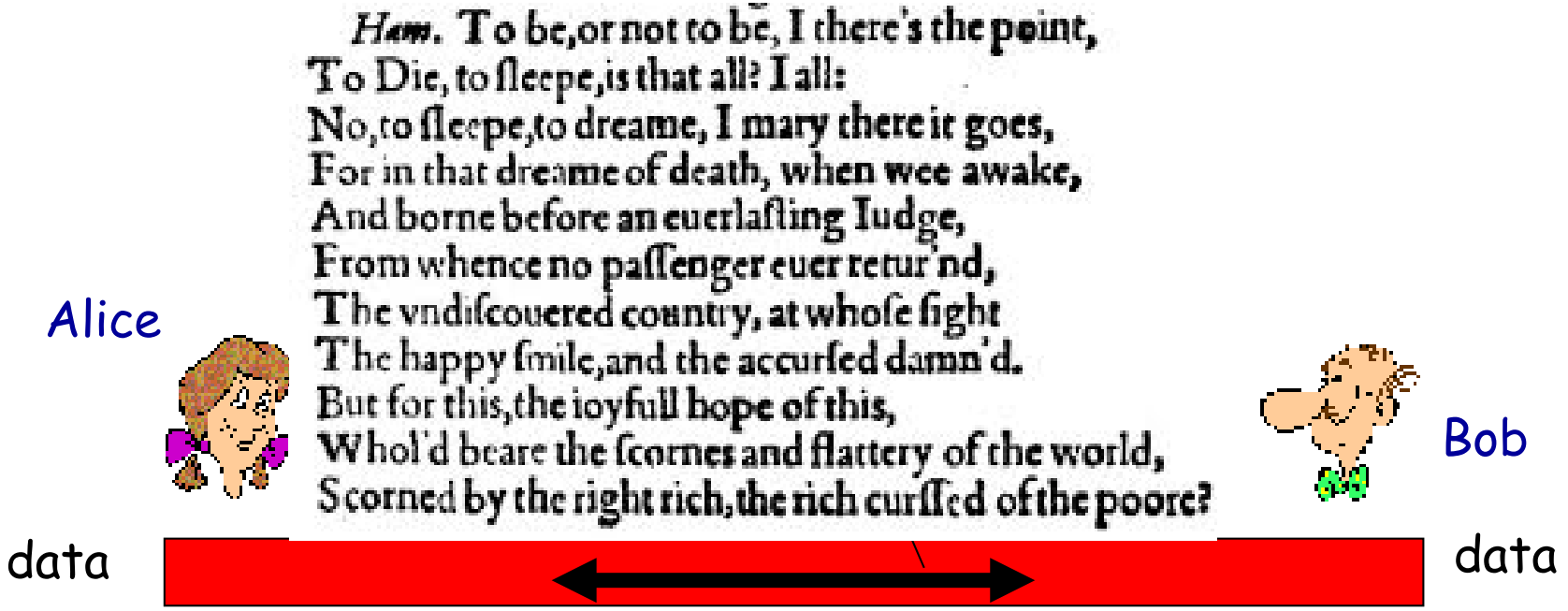
Potential Actions by Alice and Bob

- Confidentiality: only sender, intended receiver should "understand" message contents



Potential Actions by Alice and Bob

- Stealth: only sender and the intended receiver know that a hidden message is being sent



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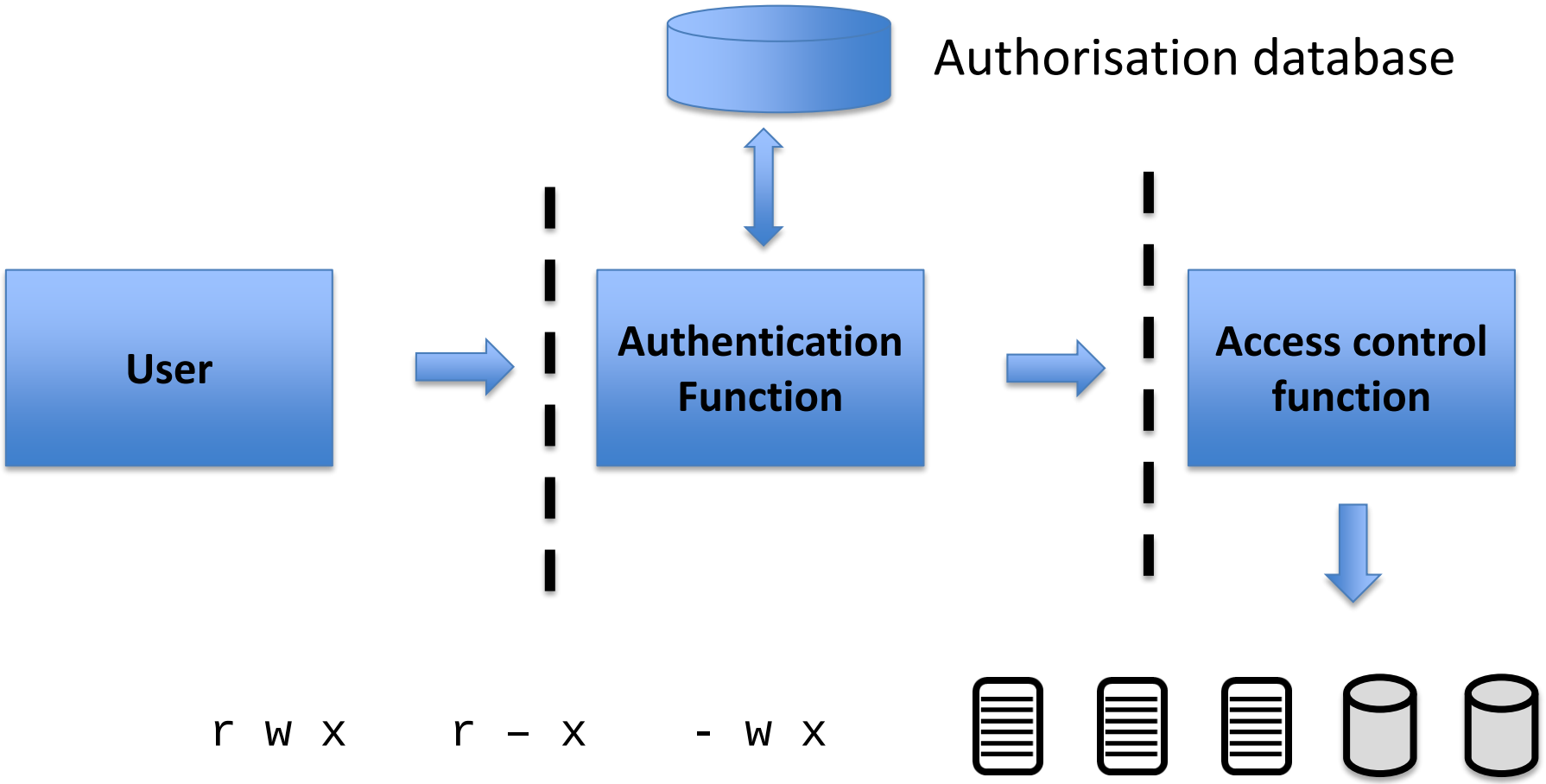
Authentication

- Authentication is a **fundamental** building block of computer security and a primary line of defence
- It provides the basis for access control and user accountability
- RFC (request for comments) 2828 defines user authentication as:
  - *The process of verifying an identity claimed by or for a system entity.*

Authentication Process

- Two broad steps in the authentication process:
  - **Identification** step presenting an identifier to the security system
  - **Verification** step presenting or generating authentication information that corroborates the binding between the entity and the identifier

Authentication Process



Means of Authentication

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- Three main aspects under the user's control which can be used to authenticate a user:
  - The user's **possession** of something (a key or card)
  - The user's **knowledge** of something (a user identifier and password)
  - An **attribute** of the user (fingerprint, retina pattern, or signature)

Password Authentication

- Widely used line of defence against intruders
- User provides name/ID and password
- System compares password with the one stored for that specified ID
- Once authenticated, the user ID:
  - determines if the user is authorized to access the system at all
  - determines the user's privileges (access rights) if so

Password Vulnerabilities

- Offline dictionary attack
- Specific account attack
- Popular password attack
- Workstation hijacking
- Exploiting user mistakes
- Exploiting multiple password use
- Electronic monitoring

Countermeasures

- Controls to prevent unauthorized access to password file
- Intrusion detection measures
- Rapid reissuance of compromised passwords
- Account lockout mechanisms
- Policies to inhibit users from selecting common passwords
- Training in and enforcement of password policies
- Automatic workstation logout
- Policies against similar passwords on network devices

Password Selection Techniques

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- **User education**
  - Users can be told the importance of using hard to guess passwords and can be provided with guidelines for selecting strong passwords
- **Computer generated passwords**
  - Users have trouble remembering them
- **Reactive password checking**
  - System periodically runs its own password cracker to find guessable passwords
- **Proactive password checking**
  - User is allowed to select their own password, however the system checks to see if the password is allowable, and if not, rejects it
  - Goal is to eliminate guessable passwords while allowing the user to select a password that is memorable

Proactive Password Checking

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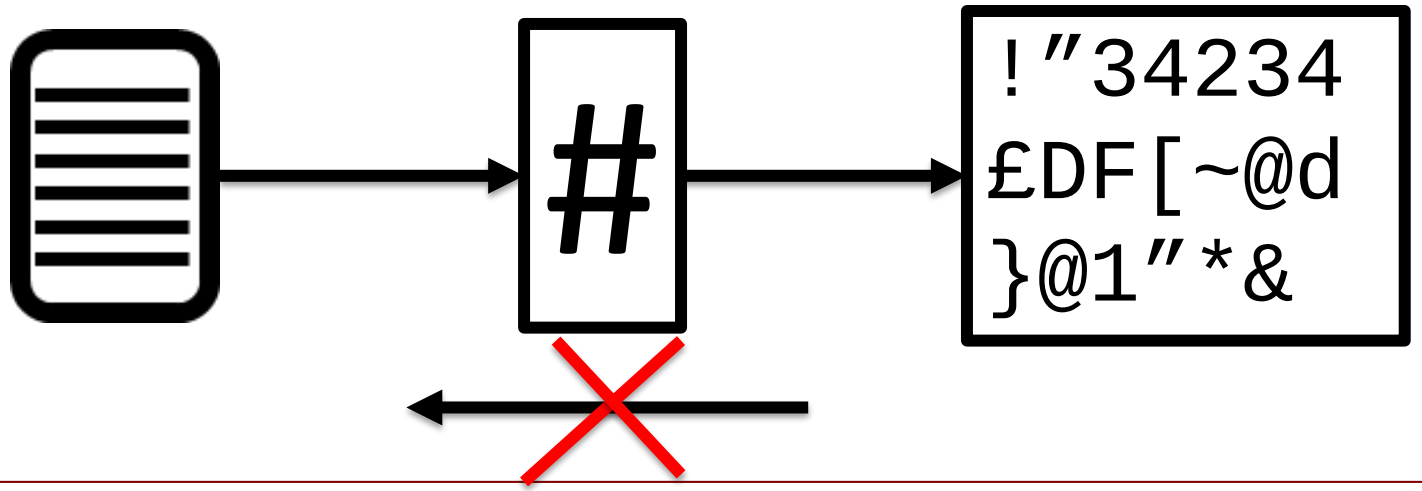
- Rule enforcement
  - Specific rules that passwords must adhere to
  - But if attackers know the rules they can optimize their attacks by only trying valid passwords
- Password "naughty list"
  - Compile a large dictionary of passwords not to use
  - Drawback: to be useful this must be very large
  - Occupies a big chunk of memory
  - Slow to search (slow login, waste of CPU cycles)

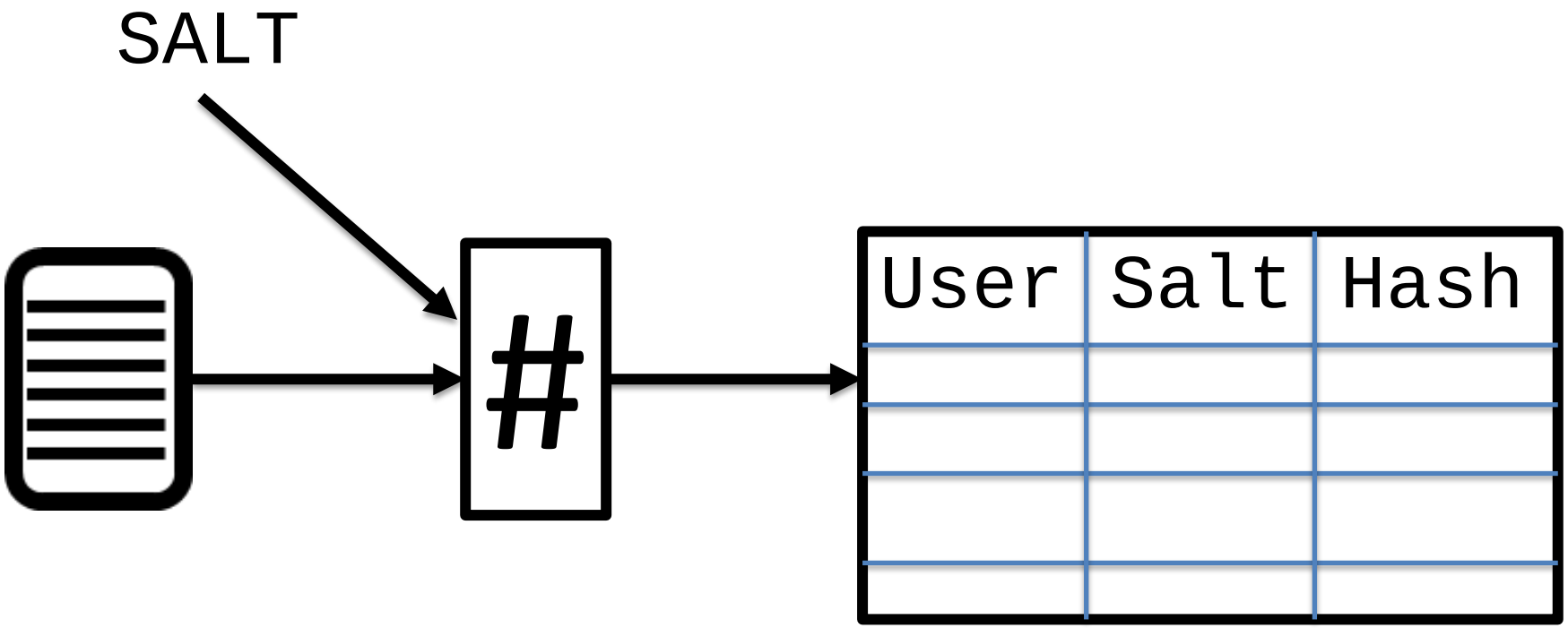
Hashed Passwords

- A hash function will convert an input sequence (can be plain text of arbitrary length, the message) to another sequence, normally of fixed length (the hash)
- It should be a one-way function, that is, a function which is practically infeasible to invert



Hashed Passwords

- Extra elements: Salt and Slow



Password implementation

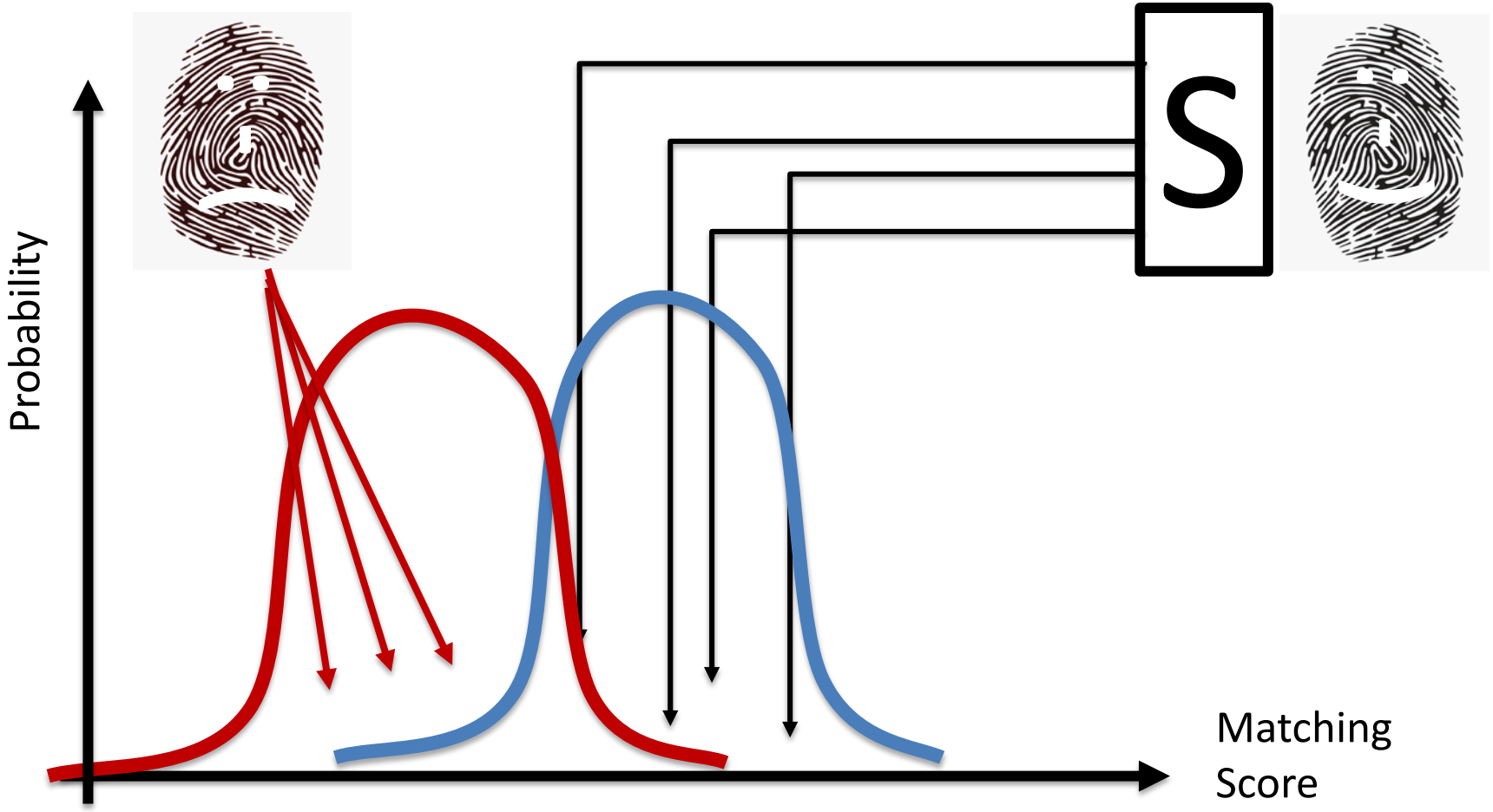
- **Unix original implementation**
- Eight ASCII characters
- Add 12 bit salt
- Encrypt 25 times (slow down)
- Translate to an 11 character sequence

- **Improved implementations**
- MD5 -> Blowfish block
- Salt 48 ->128 bits
- Password length unlimited
- 128 -> 192 hash

Alternatives to passwords

- Attempts to authenticate an individual based on unique physical characteristics
- Based on pattern recognition
- Is technically complex and expensive when compared to passwords and tokens
- Physical characteristics used include:
  - Facial characteristics,
  - Fingerprints,
  - Hand geometry,
  - Retinal pattern,
  - Iris,
  - Signature,
  - Voice, …

Issues of biometrics



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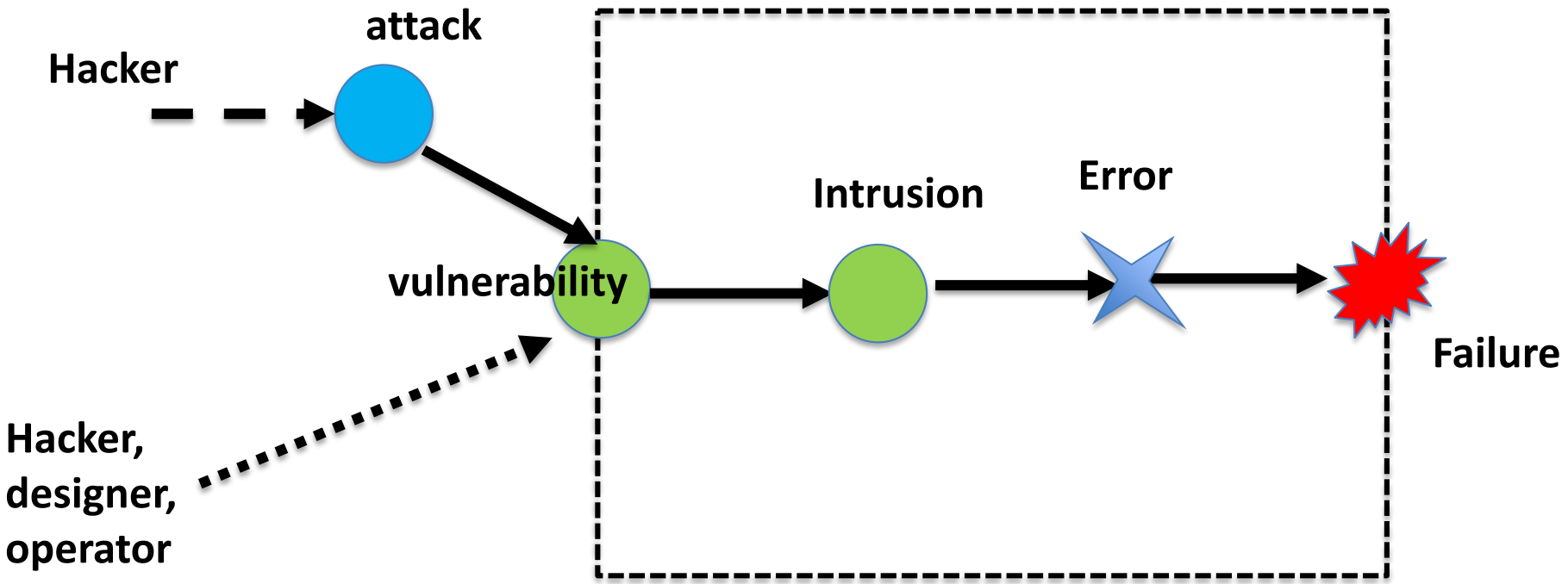
Threats

- Threats can be of various types and in different locations, to understand them better the following terminology is adopted from an EU project:
  - Malicious-and Accidental-Fault Tolerance for Internet Applications (MAFTIA)
  - http://research.cs.ncl.ac.uk/cabernet/www.laas.research.ec.org/maftia/
- The deliverable D21 of MAFTIA is recommended for further reading:
  - http://research.cs.ncl.ac.uk/cabernet/www.laas.research.ec.org/maftia/deliverables/D21.pdf

Terminology

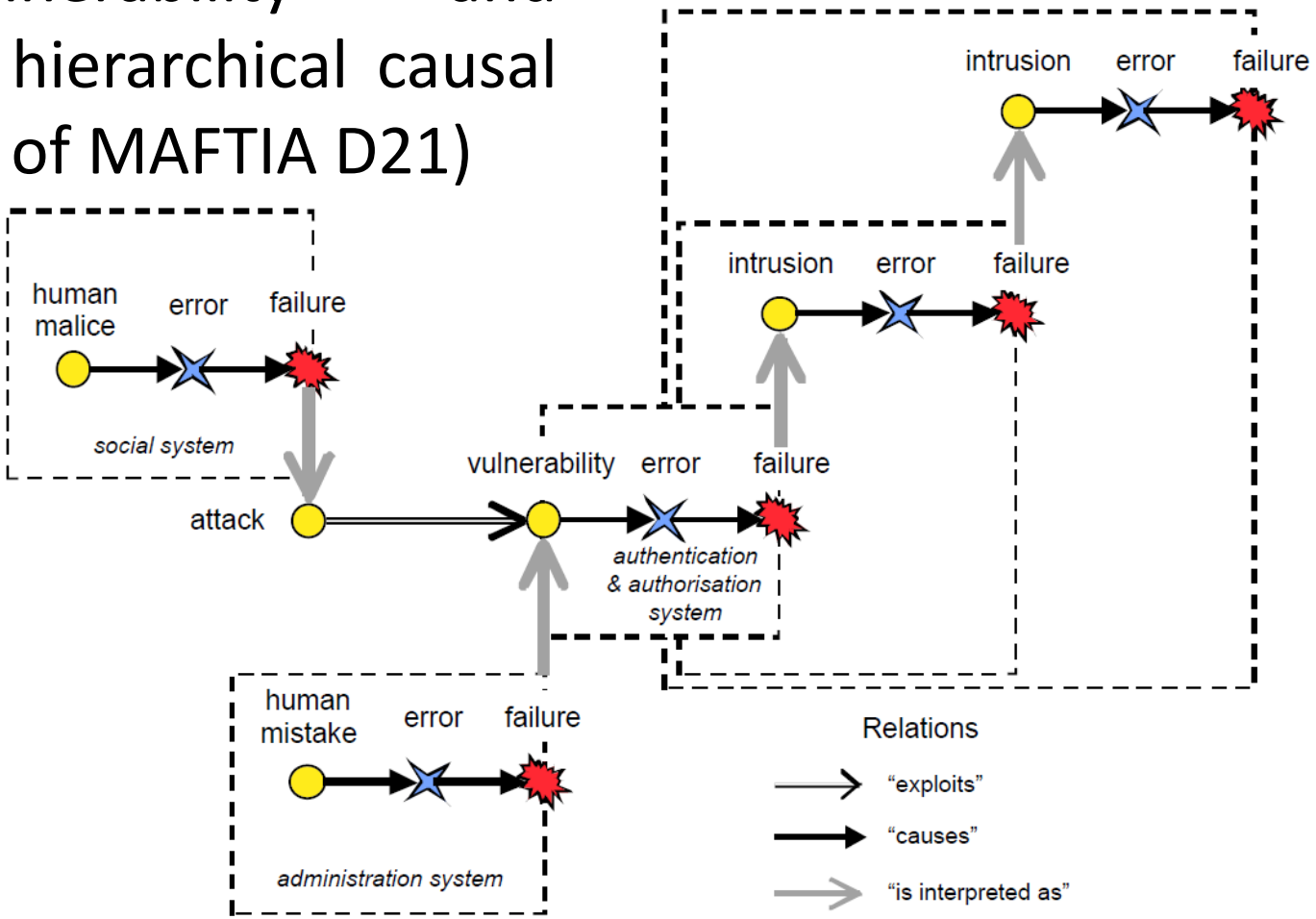
- **Fault**: adjudged or hypothesised cause of an error
- **Error**: that part of the system state that may lead to failure
- **Failure**: delivered service deviates from implementing the system function
- **Attack**: a malicious interaction *fault*, through which an attacker aims to deliberately violate one or more security properties; an *intrusion* attempt.
- **Vulnerability**: a fault created during development of the system, or during operation, that could be exploited to create an *intrusion*
- **Intrusion***:* a *malicious*, externally-induced *fault* resulting from an *attack* that has been successful in exploiting a *vulnerability*

Terminology



Hierarchical causal chain

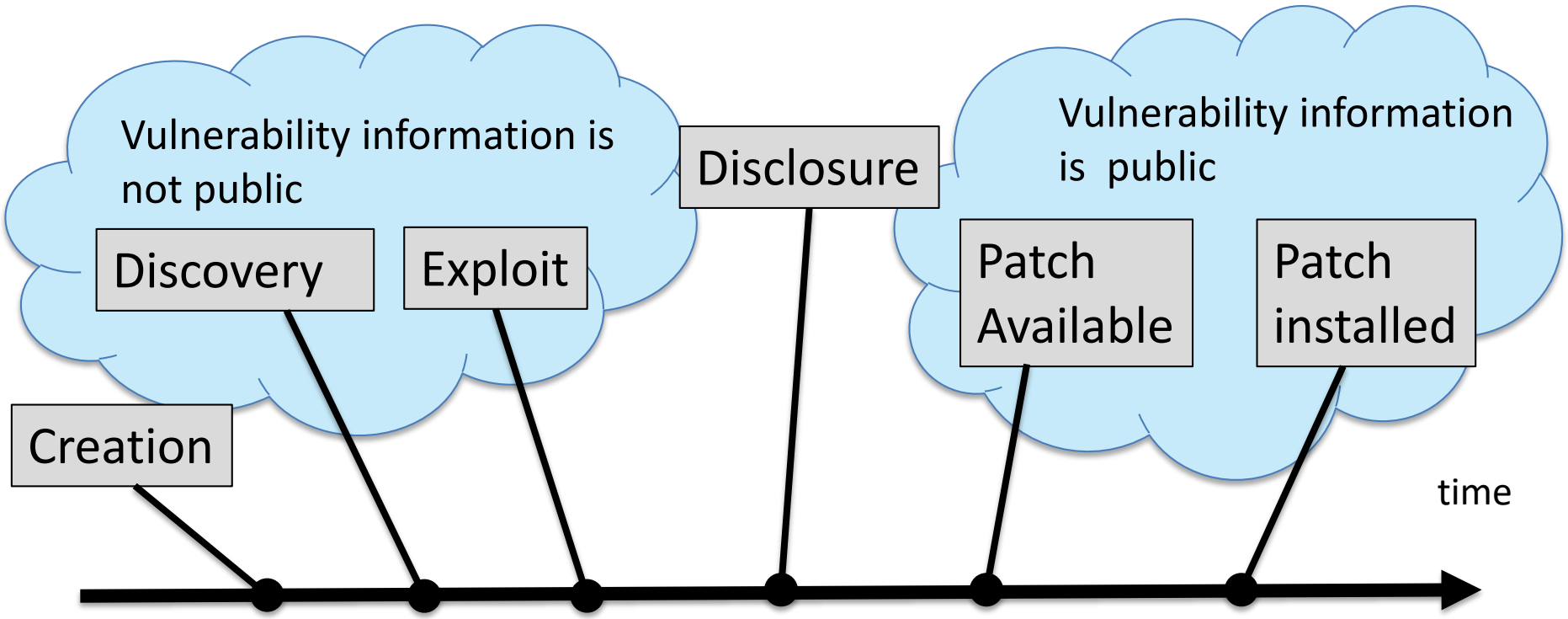
- Attack, vulnerability and intrusion in a hierarchical causal chain (figure 9 of MAFTIA D21)



Vulnerability lifecycle

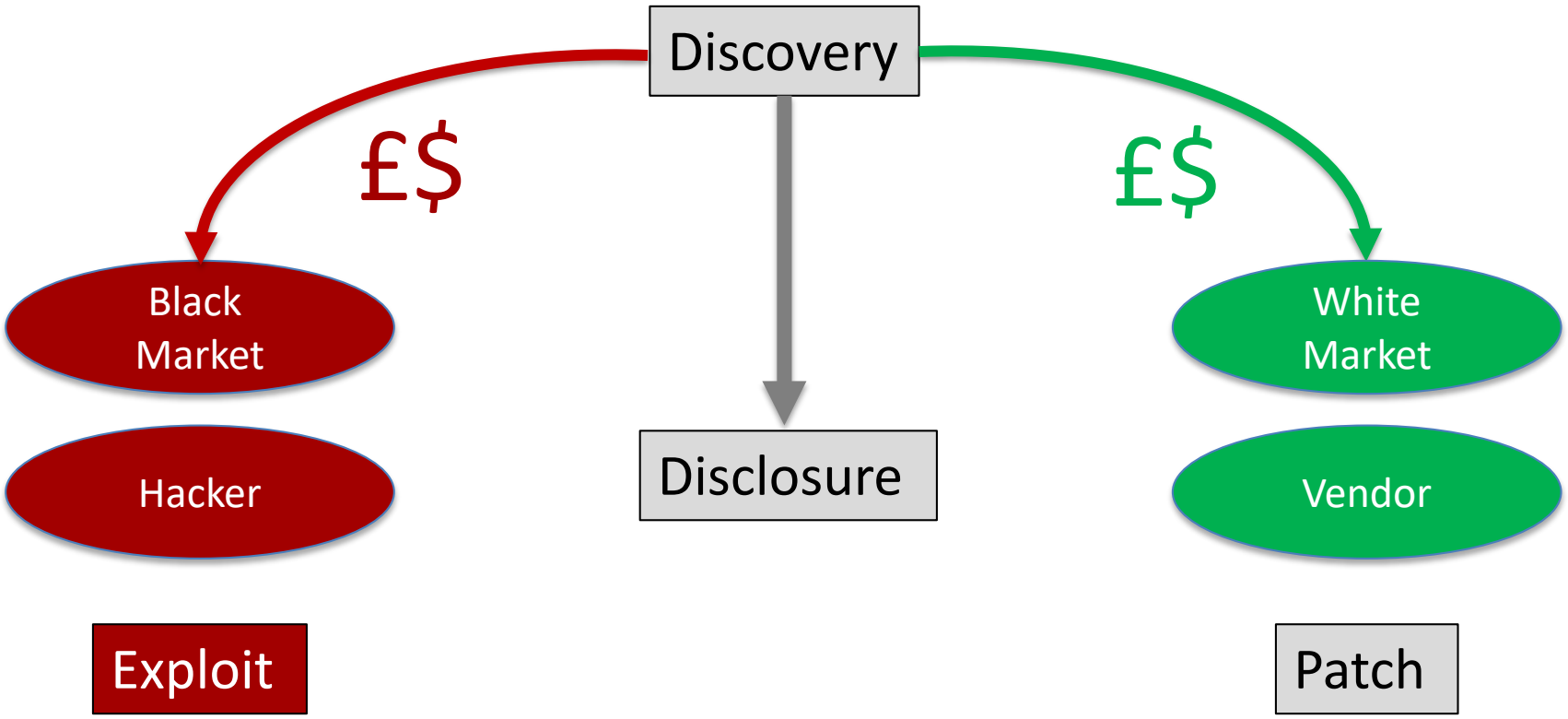
- Another useful way of conceptualising the security threats is to look at the lifecycle of a vulnerability
- A very good resource for this is the PhD thesis of Stefan Frei, from IBM Zurich. A shorter version of this work is given in the following publication (http://weis09.infosecon.net/files/103/paper103.pdf)
- Looking at the vulnerability lifecycle in this way gives us insight on the different types and stages of risks that a system may have
- It also gives us insight on the economic aspects of vulnerability disclosure and the different types of incentives that effects the decision making of both attackers and defenders

Vulnerability lifecycle



System A

Economics of a vulnerability



Historical vulnerabilities

- https://www.cvedetails.com/top-50-products.php?year=2003

Top 50 Products By Total Number Of "Distinct" Vulnerabilities in 2000

Go to year: 1999 2000 2

	Product Na
1	Linux
2	Windows 2000
3	Windows Nt
4	Freebsd
5	Internet Information
6	Hp-ux
7	IE
8	Mandrake Linux
9	Suse Linux
10	Internet Information
11	Debian Linux

Top 50 Products By Total Number Of "Distinct" Vulnerabilities in 2006

Go to year: 1999 2000

	Product N
1	Mac Os X
2	Firefox
3	IE
4	Linux Kernel
5	Seamonkey
6	Thunderbird
7	Database Server
8	Windows Xp
9	Mac Os X Server
10	Windows 2003 Ser
11	Solaris

Top 50 Products By Total Number Of "Distinct" Vulnerabilities in 2015

Go to year: 1999 2000 2

	Prod
1	Mac Os X
2	Iphone Os
3	Flash Player
4	Opensuse
5	Ubuntu Linux
6	Air Sdk
7	AIR
8	Air Sdk & Compiler
9	Debian Linux
10	Internet Explorer
11	Chrome

Top 50 Products By Total Number Of "Distinct" Vulnerabilities in 2019

Go to year: 1999 2000 2001 2002 2003 2004 2005 2006 2007 2008 2009 2010 2011 2012 20

	Product Name	Vendor Name	Product Type	Number of Vulnerabilities
1	Android	Google	OS	414
2	Debian Linux	Debian	OS	360
3	Windows Server 2016	Microsoft	OS	357
4	Windows 10	Microsoft	OS	357
5	Windows Server 2019	Microsoft	OS	351
6	Acrobat Reader Dc	Adobe	Application	342
7	Acrobat Dc	Adobe	Application	342
8	Cpanel	Cpanel	Application	321
9	Windows 7	Microsoft	OS	250
10	Windows Server 2008	Microsoft	OS	248
11	Windows Server 2012	Microsoft	OS	246

Historical vulnerabilities

- https://www.cvedetails.com/top-50-vendors.php
- https://www.cvedetails.com/top-50-vendor-cvssscore-distribution.php

Top 50 Vendors By Total Number Of "Distinct" Vulnerabilities

Go to year: 1999 2000 2001 2002 2003 2004 2005 2006 2007 2008 2009 2010 2011 2012 2013 2(

	Vendor Name	Number of Products	Number of Vulnerabilities	#Vulnerabilities/#Products
1	Microsoft	529	6814	13
2	Oracle	644	6115	9
3	IBM	1064	4679	4
4	Google	84	4572	54
5	Apple	119	4512	38
6	Cisco	3676	4167	1
7	Adobe	132	3314	25
8	Debian	97	3197	33
9	Redhat	301	2805	9
10	Linux	17	2370	139
11	Mozilla	24	2199	92
12	Canonical	30	2025	68
13	HP	3579	1794	1
14	SUN	204	1628	8
15	Opensuse	25	1315	53
16	Apache	198	1218	6
17	Fedoraproject	17	757	45
18	GNU	101	738	7
19	Novell	118	665	6
20	PHP	22	626	28
21	Qualcomm	299	602	2
22	Wireshark	1	576	576
23	SAP	226	565	3
24	Imagemagick	3	546	182
25	Huawei	1177	536	0

CVSS Score Distribution For Top 50 Vendors By Total Number Of "Distinct" Vulnerabilitie

	Vendor Name	Number of Total Vulnerabilities	# Of Vulnerabilities										Weighted Average
			0-1	1-2	2-3	3-4	4-5	5-6	6-7	7-8	8-9	9+	
1	Microsoft	6814	2	103	388	111	1016	805	389	1730	30	2240	7.50
2	Oracle	6115	2	115	272	463	1798	1670	681	518	25	571	6.00
3	IBM	4679	2	65	295	784	1263	829	460	563	30	388	5.80
4	Google	4572		23	119	18	1057	514	651	1084	31	1075	7.20
5	Apple	4512	1	53	277	44	781	544	1128	700	15	969	7.00
6	Cisco	4167	1	5	69	111	816	933	569	1169	44	450	6.90
7	Adobe	3314			18	3	409	328	176	180	1	2199	8.70
8	Debian	3197		31	130	76	883	618	620	684	7	148	6.30
9	Redhat	2805		54	212	117	693	525	437	542	9	216	6.20
10	Linux	2370	1	93	346	54	738	145	186	665	7	135	5.90
11	Mozilla	2199		9	79	12	438	453	262	388	1	557	7.20
12	Canonical	2025		30	108	58	610	387	334	393	4	101	6.20
13	HP	1794	1	11	63	43	288	250	148	406	26	558	7.50
14	SUN	1628	3	26	105	45	311	283	119	421	4	311	6.80
15	Opensuse	1315		19	66	56	291	296	253	209	3	122	6.40
16	Apache	1218		7	40	31	320	399	138	217	2	64	6.20
17	Fedoraproject	757		8	38	25	185	204	118	150	1	28	6.20
18	GNU	738	1	12	45	29	196	167	126	128		34	6.10
19	Novell	665	1	7	27	9	152	145	46	126		152	6.90
20	PHP	626		1	21	6	68	190	88	210	1	41	6.80
21	Qualcomm	602			19	3	72	70	19	164	10	245	8.00
22	Wireshark	576			24	32	184	261	7	46	3	19	5.80
23	SAP	565		3	13	26	141	171	67	98	1	45	6.40
24	Imagemagick	546			2	1	297	45	106	87		8	6.00
25	Huawei	536		3	46	14	127	106	67	113	10	50	6.40

Program Threats

- Many variations, many names
- Trojan Horse
  - Code segment that misuses its environment
  - Exploits mechanisms for allowing programs written by users to be executed by other users
  - Spyware, pop-up browser windows, covert channels
  - Up to 80% of spam delivered by spyware-infected systems
- Trap Door
  - Specific user identifier or password that circumvents normal security procedures
  - Could be included in a compiler
  - How to detect them?

Program Threats (Cont.)

- Logic Bomb
  - Program that initiates a security incident under certain circumstances
- Stack and Buffer Overflow
  - Exploits a bug in a program (overflow either the stack or memory buffers)
  - Failure to check bounds on inputs, arguments
  - Write past arguments on the stack into the return address on stack
  - When routine returns from call, returns to hacked address
    - Pointed to code loaded onto stack that executes malicious code
  - Unauthorized user or privilege escalation

Program Threats (Cont.)

- Viruses
  - Code fragment embedded in legitimate program
  - Self-replicating, designed to infect other computers
  - Very specific to CPU architecture, operating system, applications
  - Usually borne via email or as a macro

Program Threats (Cont.)

- Virus dropper inserts virus onto the system
- Many categories of viruses, literally many thousands of viruses
  - File / parasitic
  - Boot / memory
  - Macro
  - Source code
  - Polymorphic to avoid having a virus signature
  - Encrypted
  - Stealth
  - Tunneling

System and Network Threats (Cont.)

- Port scanning
  - Automated attempt to connect to a range of ports on one or a range of IP addresses
  - Detection of answering service protocol
  - Detection of OS and version running on system
  - nmap scans all ports in a given IP range for a response
  - nessus has a database of protocols and bugs (and exploits) to apply against a system
  - Frequently launched from zombie systems
    - To decrease trace-ability

System and Network Threats (Cont.)

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- Denial of Service
  - Overload the targeted computer preventing it from doing any useful work
  - Distributed denial-of-service (DDOS) come from multiple sites at once
  - Consider the start of the IP-connection handshake (SYN)
    - How many started-connections can the OS handle?
  - Consider traffic to a web site
    - How can you tell the difference between being a target and being really popular?
  - Accidental – CS students writing bad fork() code
  - Purposeful – extortion, punishment

Countermeasure

- Involves four complementary courses of action:
- **Prevention** E.g. Secure encryption algorithms / Prevent unauthorized access to encryption keys
- **Detection** E.g. Intrusion detection systems / Detection of denial of service attacks
- **Recovery** E.g. Use of backup systems
- **Response** E.g. Upon detection, being able to halt an attack and prevent further damage

More information about security threats, vulnerabilities

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- https://us-cert.cisa.gov/
- https://www.ncsc.gov.uk/

- https://www.cvedetails.com/

- https://www.ibm.com/security/xforce
- https://www.broadcom.com/products/cyber-security (Symantec)
- https://www.broadcom.com/support/security-center/publications/archive
- http://cve.mitre.org/

More information about security threats, vulnerabilities

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- Buffer overflows
  - http://www.youtube.com/watch?v=8xonDJe3YxI
- Hire the hackers:
  - http://www.ted.com/talks/misha_glenny_hire_the_hackers.html
- Check out other talks in the series “Hire the hackers”:
  - http://www.ted.com/playlists/10/who_are_the_hackers.html

Systems Security

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- Computer Security Overview
- Problems: Security Violation Methods
- Actions to Mitigate the Problems
- Authentication, Passwords and Alternatives
- More on Threats